

Question

Trace element **X** is named after a Western European country. Extraction of **X** from minerals of elements from the same group requires a complex process, in which the key step is the formation of a precipitate **A** containing **X** and a precipitate **B** of the same type but without **X** to enrich **X**.

The element **X** can react with a colorless irritating gas **C** to form a gray semi-metallic solid **D**, and **D**-type compounds have important industrial applications.

A monovalent anion **E** containing a mass fraction of 37.11% of **X** has a C_2 axis and 4 bridging bonds, while the neutral compound **F** with the same elemental composition as **E** exhibits a +2 apparent valence for **X**, but in reality, there is an anion-cation pair. **F** dissolves in diethyl ether to obtain a solvate **G**, which also contains discrete anions and cations with the same coordination number for the anions and cations.

The reaction of $LiBH_4$ with a high-valent halide of **X** at $-45^\circ C$ yields a ternary compound **H**, where $\omega_X = 69.43\%$, which can be approximately considered to have C_{2v} symmetry.

A binary compound **I** of **X** can be prepared below $308K$. Its monomer is a viscous liquid with slight polarity, while the polymer is a white solid. Both have strong reducing properties. The 1 : 1 adduct formed by **I** with ligand **J** can be used to prepare **I** and can also form a 1 : 2 complex **K**, whose configuration is different from **H**.

C, J, D contain the same element, and **J** can be prepared by complete methylation of **C**.

Given the following options for the above process, let x be the product of the serial numbers of all incorrect statements, and y be the sum of the serial numbers of all correct statements. Find $\frac{\sqrt{x}}{y+1}$ and select the option with an absolute error < 0.001 , if there is no option that meets the condition, choose **A**.

1. The acidity of precipitate **A** is stronger than that of precipitate **B** partly due to the effect of electron shielding on the effective nuclear charge
2. Gas **C** dissolves in water to give an acidic solution
3. Compound **D** reacts violently with water at room temperature

4. One C_2 axis of anion **E** passes through a hexacoordinate atom and two ligands
5. Compound **F** has a molecular weight of 281.2
6. The anion and cation of solvate **G** have the same stereoconfiguration
7. One molecule of compound **H** contains 13 atoms
8. Compound **I** is extremely prone to decompose into its elemental form at room temperature
9. The number of atoms of each element in ligand **J** constitutes a geometric sequence
10. The configuration of complex **K** is square pyramidal

A. All other options are incorrect

B. 0.373

C. 3.162

D. 0.727

E. 3.036

F. 1.793

G. 1.623

Answer

Correct Answer: **F**

Detailed Explanation

X is both a rare element and named after a Western European country. Consider the three elements *Ga*(Gallia), *Ge*(Germania), *Sc*(Scandia).

Consider the reaction of $LiBH_4$ with a high-valent halide of **X** to obtain a ternary compound **H**. The strong reducing power of $LiBH_4$ completely replaces the halogen and provides two kinds of ions, BH_4^- , H^- . Therefore, the other two elements besides **X** should be *B*, *H*. Let the chemical formula of **H** be $XH_x(BH_4)_y$, then $\omega_X = \frac{Ar(X)}{Ar(X) + 1.008x + 14.84y} = 69.43\%$.

Consider the element **X** = *Ga*, substitute to get the equation $1.008x + 14.84y = 30.70$, the approximate non-negative integer solution is $x = 1, y = 2$, corresponding to **H** = $GaH(BH_4)_2$, actually $\omega_{Ga} = \frac{69.72}{100.408} = 69.43\%$.

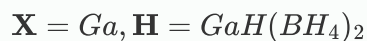
Consider the element **X** = *Ge*, substitute to get the equation $1.008x + 14.84y = 31.97$, the approximate non-negative integer solution is $x = 2, y = 2$, corresponding to **H** = $GeH_2(BH_4)_2$, actually $\omega_{Ge} = \frac{72.61}{104.306} = 69.61\%$.

Consider the element **X** = *Sc*, substitute to get the equation $1.008x + 14.84y = 19.80$, the approximate non-negative integer solution is $x = 5, y = 1$, corresponding to **H** = $ScH_5(BH_4)$, actually $\omega_{Sc} = \frac{44.96}{64.84} = 69.34\%$.

Therefore, **X** = *Ga*, **H** = $GaH(BH_4)_2$, approximately has C_{2v} symmetry. The number of atoms in one molecule of compound **H** is 12, statement 7 is wrong.

CHECKPOINT

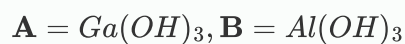
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The homologous mineral is bauxite, then the two precipitates are $\mathbf{A} = Ga(OH)_3, \mathbf{B} = Al(OH)_3$. Acidity $Ga(OH)_3 > Al(OH)_3$, because the poor shielding effect of 3d electrons and the lanthanide contraction make the effective nuclear charge of Ga larger, and the ion potential of Ga^{3+} is greater than Al^{3+} , statement 1 is correct.

CHECKPOINT

1 PTS



The strong reducing binary compound $\mathbf{I} = GaH_3$ is similar to BH_3 , and the corresponding polymer $(GaH_3)_n$ is a white solid. The decomposition reaction $2GaH_3 \rightarrow 2Ga + 3H_2$ proceeds rapidly above $243K$, statement 8 is correct.

CHECKPOINT

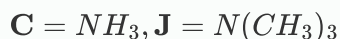
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The colorless irritating gas $\mathbf{C}$ is considered HCl, HBr, NH_3 , etc., because the ligand $\mathbf{J}$ can be prepared by complete methylation reaction of $\mathbf{C}$, consider $\mathbf{C} = NH_3, \mathbf{J} = N(CH_3)_3$.

CHECKPOINT

1 PTS

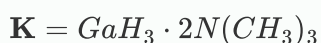


The gas **C** is dissolved in water to obtain weakly alkaline ammonia water, statement 2 is wrong. The number of element atoms in **J** constitutes a geometric sequence of 1 : 3 : 9, statement 9 is correct.

The formed complex of **I** : **J** = 1 : 2 is $\mathbf{K} = \text{GaH}_3 \cdot 2\text{N}(\text{CH}_3)_3$. The configuration is trigonal bipyramidal, and the two large ligands tend to be in the axial direction, statement 10 is wrong.

CHECKPOINT

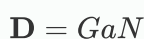
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Therefore, the gray semi-metallic solid $\mathbf{D} = \text{GaN}$ produced by **X** and **C** is widely used in industry as a semiconductor, and it is not easily soluble in water at room temperature and does not react violently with water, statement 3 is wrong.

CHECKPOINT

0.5 PTS

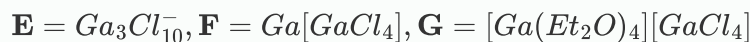


The common valence of *Ga* is +1, +3, while the apparent valence in **F** is +2, indicating that the two common valences account for half each; the monovalent anion **E** with the same element composition has a C_2 axis and 4 bridging bonds, inferring that both contain halogen. Let the chemical formula of **E** be GaY_a^- , then $\omega_{\text{Ga}} = \frac{Ar(\text{Ga})}{Ar(\text{Ga}) + aAr(\text{Y})} = 37.11\%$, simplified to $aAr(\text{Y}) = 118.15$, there are solutions 6.22*F* · 3.33*Cl* · 1.48*Br* · 0.93*I* respectively.

Combined with common valences, we get $\mathbf{E} = \text{Ga}_3\text{Cl}_{10}^-$. The corresponding $\mathbf{F} = \text{Ga}[\text{GaCl}_4]$, the solvate of $\mathbf{G} = [\text{Ga}(\text{Et}_2\text{O})_4][\text{GaCl}_4]$ dissolved in ether, ensures that the coordination number of anions and cations is the same.

CHECKPOINT

1.5 PTS



Therefore, the C_2 axis of $\mathbf{E}$ passes through a central six-coordinated Ga atom and the upper and lower two Cl ligands, statement 4 is correct.

The molecular weight of $\mathbf{F}$ is $2 \times 69.72 + 4 \times 35.45 = 281.24$, statement 5 is correct.

The anion of $\mathbf{G}$ has four-coordinated T_d symmetry, and the cation has seesaw-shaped C_{2v} symmetry, the stereoconfigurations are different, statement 6 is wrong.

The correct statements are 1, 4, 5, 8, 9, and the wrong statements are 2, 3, 6, 7, 10.

CHECKPOINT

1 PTS

正确的叙述为 1, 4, 5, 8, 9, 错误的叙述为 2, 3, 6, 7, 10

Calculate $x = 2 \times 3 \times 6 \times 7 \times 10 = 2520$, $y = 27$, $\frac{\sqrt{x}}{y+1} = 1.793$.

CHECKPOINT

1 PTS

$$x = 2520, y = 27, \frac{\sqrt{x}}{y+1} = 1.793$$